



**CSE370: Database Management System
Assignment on Functional Requirements
Proposed Project Title: Project Title**

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Project Overview

The **Disaster Relief Management System** is a platform designed for efficient management of relief resources, volunteers, donors and disaster zones. This system enables administrators to track warehouse inventories, relief shipments, oversee the areas which are affected by disasters and supervise volunteers. It keeps a record of relevant data for each disaster zone, which includes location, severity and current state, so that admins can allocate volunteers and resources more efficiently. It also supports management of donors by keeping records of them. Donors can be of three distinct types- NGO, individual and corporate donors. Corporate donors are recognized by tax ID and a tier based ranking system.

Moreover the platform provides secure authentication for all users, who are either admins or customers, ensuring that all the users are recognized by the platform and that they can access all the relevant features. The system also enables automated inventory monitoring along with low-stock alerts and expiry date review that help maintain the availability and quality of the relief supplies. Additionally, volunteers are tracked to be matched and deployed, depending on their availability, to improve the efficiency of emergency response operations. Overall, the system aims to improve coordination and timely delivery of aid and supplies during disasters.

Tech Stack:

- · Language:
- · Frontend Framework:.js
- · Backend Framework:
- · Styling:
- · Database: MySQL
- · ODM:
- · Deployment:
- · External APIs:

Functional Requirements

Module 1	
Member	Feature Description
1	Login/Logout System: Registered users can log in using valid credentials to access the system. The system records login and logout timestamps, IP addresses, and securely manages user sessions to ensure account security.
2	User Registration & Role-Based Access Control: New users can create an account by providing their name, password, and other required information. The system stores user details securely and assigns the appropriate role (Admin or Customer), allowing users to access the Disaster Relief Management System. The system identifies the logged-in user's role and grants appropriate permissions. Users can access relevant features depending on their role; admins can supervise inventory management.
3	User Authentication & Session Management: The system validates user credentials, maintains active sessions, and prevents unauthorized access to protected system resources, ensuring secure and uninterrupted access for authenticated users.

Module 2	
Member	Feature Description
1	Centralized Inventory Stock Monitor & Automated Low-Stock Alert: Administrators can monitor inventory levels across all warehouses from a centralized dashboard. The system displays available quantities, item categories, and warehouse locations to simplify inventory management. It monitors inventory stock and automatically identifies items that need restocking after the quantity falls below a certain threshold.

2	Disaster Zone & Relief Shipment Tracking: The system records and tracks the shipment of relief supplies from warehouses to disaster zones. Dispatch timestamps and shipment details are maintained to ensure timely delivery of essential resources. Admins can create, update, and monitor disaster zones by maintaining information such as location, severity level, and current status.
3	Expiry Date Auditor: The system monitors the expiration dates of relief supplies and calculates the remaining days before expiry. Expiring items are flagged so they can be prioritized for distribution or replaced before becoming unusable.

Module 3	
Member	Feature Description
1	Emergency Volunteer Matchmaker & Volunteer Deployment Tracker: The system matches volunteers to disaster zones based on their skills, specialties and availability. Administrators can assign volunteers to disaster zones and monitor their current role, deployment history, and hours contributed, ensuring efficient coordination during response operations.
2	Donor Management & Corporate Donor Rewards and Ranking System: The system manages corporate donor information, including Tax ID, Company Size, and Tier Ranking. Organizations are ranked based on their contribution level, allowing administrators to recognize and encourage continued support.
3	Volunteer Contribution Monitoring: The system tracks volunteer participation, deployment history, and hours contributed across disaster relief operations. Administrators can review volunteer activity to improve workforce planning and future deployment decisions.